

Ahmed Ikram | Data Engineering & Analytics

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PROFILE

4th-year BSc Computer Science (Data Science & AI) student at the University of Dundee, on track for a First-Class degree. Specialises in event-driven pipeline engineering and cloud lakehouse architecture - built a Kafka KRaft pipeline sustaining 930K+ events at 100+ msg/sec with exactly-once semantics on AWS, and a Databricks DLT medallion pipeline (153,380 rows, 0 dropped across 7 quality gates) with dual-dialect dbt models and Terraform-provisioned infrastructure, validated across 2,015 tests in total. Seeking a graduate Data Engineering or Analytics role in a data-first team.

EDUCATION

High School of Dundee

Aug 2017 - June 2023

- National 5 Computer Science, Maths, English, Economics, Physics, Chemistry, Biology (A)
- Higher Computer Science, Maths and English (A), Higher Chemistry (B) and Higher Physics (C)
- Advanced Higher Computer Science (C)

University of Dundee - BSc (Hons) Computer Science (Data Science and AI)

Sep 2023 - Present

- Expected Graduation: June 2027

PROJECTS

W3C Web Logs ETL Pipeline | *Apache Airflow, Python, Databricks, dbt, Apache Spark, PySpark, Azure SQL, Delta Lake, Terraform, Docker, Power BI, DAX, Power Automate, Grafana, Prometheus* | Feb 2026 - June 2026

- Ingested 93 W3C IIS log files (155,380 rows, 3 dropped) on Azure Databricks DLT via Auto Loader with a custom parser detecting 14- vs 18-field IIS formats at parse time; Silver enriches every row via a consolidated MaxMind GeoIP struct UDF - 3.5× faster than 7 separate UDFs - achieving 99.99% known-country coverage across 153,377 enriched, 31-column rows across 7 DLT quality expectations
- Cut Silver→Azure SQL export 8-9× (413s → ~45s) via Spark-side pre-filtering before collect(), tuple(row) serialisation over asDict() (4.7M fewer dict allocations) and removing redundant .count() scans; dimension tables (geolocation & user-agent) built via SQL MERGE upsert with SHA-256 hash natural keys and -1 sentinel rows preserving FK integrity
- Built 16 dbt models (10 staging + 6 marts) compiling against both PostgreSQL and T-SQL from a single codebase via inline Jinja branches and 18 compatibility macros; incremental fact table dedupes on a 12-component MD5 key, catching 213 duplicate rows across 155K+ raw records; 121 dbt tests (not_null, unique, accepted_values, relationships, expression_is_true) enforce referential integrity across all 9 dimensions before data reaches Power BI
- Orchestrated via 4 Airflow DAGs with Dataset-driven decoupling - no ExternalTaskSensor, no hard-coded DAG IDs; provisioned 30+ Azure resources across 2 Terraform configurations with a fully OIDC-federated GitHub→Azure auth chain and zero static credentials; 613 tests across 5 tiers gate every push; 3 Grafana dashboards (23 panels) fed by a custom freshness probe with 10 combined Prometheus/Azure Monitor alert rules routing to a Slack notification channel

ATM Log Aggregation, Analysis & Diagnostics Platform | *FastAPI, Apache Kafka, PostgreSQL, Redis, MLflow, SageMaker, Docker, Terraform, AWS* | Mar 2026 - June 2026

- Backend and data engineering lead on a 7-person Agile team for NCR Atleos - independently replaced the direct-DB generator with a full event-driven Apache Kafka KRaft ingestion pipeline processing 2.5M+ events every day at 100+ msg/sec across 7 simultaneous log sources into 2 topics (3 partitions each); 7 source-specific parsers with .get() safe defaults ensure schema drift never halts ingestion; malformed records route to a dead-letter table; hybrid Redis SET + 1h TTL primary and 10K LRU fallback deduplication guarantees exactly-once semantics across consumer restarts; failed messages retry via a Redis Stream DLQ with exponential backoff (5s → 10s → 20s, max 3 retries before exhausted)
- Designed a PostgreSQL data lake with JSONB payloads (new log sources require only a new parser - zero schema migrations), a v_unified_analysis UNION ALL view with COALESCE field normalisations as the single ML query target, and 14 composite B-tree indexes on (timestamp, entity, type) axes sustaining sub 100ms queries at 2M+ rows; ThreadedConnectionPool (min=5, max=50) with 3-retry backoff serves concurrent API, detection, and cleanup threads; batched retention cleanup (5,000 rows/batch, is_active=1 filter) preserves unresolved anomalies regardless of age
- Tracked all training runs and 7 model artifacts in MLflow (RDS PostgreSQL backend + S3 artifact store) with champion aliases and auto-retrain on startup if artifacts are missing or version-skewed; deployed XGBoost to a production SageMaker endpoint (laad-xgb-champion, ml.t2.medium) for live cross-check inference; full AWS estate provisioned via Terraform (10 modules, 118 resources) including multi-AZ VPC, ECS Fargate, RDS, S3, CloudFront, Secrets Manager (8 secrets), and 6 least-privilege IAM roles; validated via 1,402 tests catching silent data loss under 50 concurrent writes and exactly-once semantics end-to-end

Haggis Species Classification & Predictive Modelling | Python, Scikit-learn, XGBoost, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook | Nov 2025 - Dec 2025

- Ran a complete ML pipeline from EDA through feature engineering to model evaluation; engineered and validated 2 novel ratio-based features using sklearn pipelines (ColumnTransformer, RobustScaler) that ranked among the top-3 predictors across all models, whilst preventing data leakage
- Trained and compared 7 classifiers (XGBoost, Random Forest, Logistic Regression and others) achieving ~90% accuracy and tuned hyperparameters via GridSearchCV with 5-fold cross-validation
- Applied K-Means and DBSCAN clustering and built a Linear Regression model ($R^2=0.756$) and validated assumptions via residual diagnostics and standardised coefficient analysis

Project Management Platform with GitHub Integration | Python, Flask, PostgreSQL, Alembic, React, Socket.io, AWS, Docker, GitHub Actions, CI/CD | Jan 2025 - Apr 2026

- Designed a 12-table PostgreSQL schema built around actual API access patterns - indexes on FK, join, and filter columns throughout; dedicated tables for audit logs (filterable by user, action, resource, and date with configurable retention and cleanup), typed saved reports (JSON summary and details fields), system settings, GitHub repository tracking, and bidirectional task-to-Issue/PR link associations enforced at the database layer
- Deployed production-like AWS data infrastructure: ECS Fargate & ALB serving the Flask API, RDS PostgreSQL in a private subnet accessible only from ECS at the network layer, and a React SPA on S3, CloudFront & ACM; rolling deploys tag each image with the commit SHA and latest for immutable rollback references at any prior build
- 1,452 automated tests at 85%/85% line coverage hard-gate every PR via 7 path-aware CI jobs (lint: ruff + ESLint, security: pip-audit + npm audit, backend-tests against real Postgres 15, frontend-tests, Cypress E2E) - all must pass before any AWS step runs; OIDC federation authenticates the pipeline to AWS without static credentials stored anywhere. Docker layer caching cuts rebuilds by 70%+. Dependabot + CodeQL on weekly cadence

Global Superstore Business Intelligence Dashboard | Excel

- Analysed 51,000 sales records across regions, product categories, and time periods; identified that aggressive discount strategies in the Furniture segment were directly suppressing net profit margins - quantified the impact and surfaced it as a prioritised KPI recommendation alongside additional category and regional findings
- Built a slicer-driven interactive dashboard enabling non-technical stakeholders to independently explore regional and category-level performance without requiring analytical support; delivered findings as a structured business recommendations document and recorded walkthrough

TECHNICAL SKILLS

Languages	Python · SQL · TypeScript · JavaScript · Bash
Pipeline Orchestration	Apache Airflow · Apache Kafka · dbt · Apache Spark, PySpark
Platforms	Databricks (DLT, Auto Loader, Unity Catalog) · Delta Lake
Databases	PostgreSQL · MongoDB · MySQL · SQLite · Redis · Alembic
Analytics & BI	Pandas · NumPy · Power BI · DAX · Matplotlib · Seaborn · Power Automate · Jupyter Notebook
Observability	Grafana · Prometheus · Azure Monitor · StatsD · Slack (alert routing)
Cloud & DevOps	AWS (ECS Fargate, EC2, S3, RDS, IAM, CloudFront, SageMaker) · Azure (Databricks, Azure SQL, Data Factory, ADLS) · Terraform · Docker · GitHub Actions · Linux
Testing	Pytest · dbt Tests · Terraform

CERTIFICATES

<u>AWS Academy Graduate - Machine Learning Foundations</u>	Feb 2025
<u>AWS Academy Graduate - Cloud Foundations</u>	Mar 2024
<u>Foundations of Azure AI: Concepts, Capabilities, and Implementation</u>	Oct 2024

EXPERIENCE

Beaully Mini Market , Dundee, Scotland	
Retail Assistant	September 2023 - Present
• Diagnosed and resolved Point-of-Sale and stock system failures rapidly to minimise service disruption	
Zulus Peri Peri Grill , Dundee, Scotland	
Front of House Worker	June 2021 - September 2023
• Trained 5 new staff members; managed competing priorities across dine-in, takeaway, and delivery orders under sustained pressure	